

USE CASE CATALOGUE

I. Actor Catalogue

Actor	Description	Note
Student	Uses shared electric vehicles, finds suitable Hubs, reserves parking spaces, and requests charging for a privately owned EV.	
Operator	Monitors the Hub network, coordinates vehicle redistribution, handles incidents, supervises charging, and evaluates hypothetical operating scenarios.	
External Telemetry Provider	Supplies vehicle, parking, and charging-point status updates. This may be an IoT gateway or a separate data-simulation service.	

II. Shared Use Cases - Student and Operator

Use case	Goal and expected outcome	Note
Sign In	Authenticate the user and establish a session with the permissions assigned to the Student or Operator role. Invalid credentials do not create a session.	
Sign Out	End the current session so that subsequent protected actions require authentication again.	

III. Student Use Cases

Hub & Identity Use Cases

Use case	Goal and expected outcome	Note
Manage Personal Profile	View and update permitted profile information, such as contact details. The system validates changes and preserves the student's identity and role.	
Find a Mobility Hub	Search or filter Hubs using criteria such as location and required services. The system returns matching Hubs for further inspection.	
View Hub Status	Inspect a selected Hub's shared-vehicle availability, parking occupancy, charging-point availability, and operational status. Display the data timestamp or freshness indicator alongside near-real-time information.	

Vehicle Rental Use Cases

Use case	Goal and expected outcome	Note
Find a Suitable Shared Vehicle	Find available vehicles using relevant criteria, such as Hub, vehicle type, and battery level. Return vehicles suitable for the intended journey.	
View Shared Vehicle Details	Inspect a selected vehicle's identifier, type, current Hub, battery level, availability, and operational condition.	
Reserve a Shared Vehicle	Request a reservation for a selected vehicle. The system rechecks availability and records the reservation without allocating the same vehicle to incompatible bookings.	Include: Check Vehicle Availability - Verify that the selected vehicle is available for the requested allocation. View Alternative Hubs extends Reserve a Shared

		Vehicle
Pick Up a Shared Vehicle	Confirm the student's entitlement to collect a vehicle and begin a rental. Recheck its availability and operational condition before marking it as in use.	Include: Check Vehicle Availability - Recheck the vehicle's current condition and availability, accounting for the student's own valid reservation where applicable.
Return a Shared Vehicle	Register the vehicle's arrival at a permitted destination Hub, confirm that return conditions are satisfied, and end the rental. Update vehicle location and parking occupancy.	Report a Vehicle Problem extends Return a Shared Vehicle
View Their Rental Reservations and Trips	Review personal vehicle reservations, an active rental, and completed trips, including their current statuses.	
Cancel a Vehicle Reservation	Cancel an eligible reservation before pickup and release the associated vehicle allocation according to the agreed cancellation policy.	
Report a Vehicle Problem	Submit a problem report identifying the vehicle and describing the issue. Create an incident record for Operator review.	Report a Vehicle Problem can be initiated independently by the Student. It also has an «extend» relationship with Return a Shared Vehicle when the Student chooses to report an issue during the return process.

Parking & Private EVs Use Cases

Use case	Goal and expected outcome	Note
Reserve a Parking Space	Select a Hub and requested parking period, provide necessary vehicle information, and submit a reservation. The system checks capacity for that period and confirms the	Include: Check Parking Availability - Check whether capacity can accommodate the requested parking interval before confirming it. View Alternative Hubs

	booking only if it can allocate the required space.	extends Reserve a Parking Space
Submit a Private-EV Charging Request	Specify the vehicle, selected Hub, expected arrival/departure times, and charging needs. The system validates and records the request for scheduling. Acceptance of the request does not automatically guarantee a charging slot.	
View My Parking and Charging Bookings	Review personal parking reservations and charging requests, including their Hub, time period, status, and assigned charging schedule where available.	
Cancel a Parking Reservation	Cancel an eligible parking reservation and release its reserved capacity. If a charging request depends on that reservation, apply the agreed dependency rule and explain the resulting changes.	
Cancel a Charging Request	Cancel an eligible pending or scheduled charging request and release any allocated charging resources. Cancellation of charging need not cancel parking.	
Check In a Private EV	Confirm the vehicle's arrival against a valid parking reservation and start the parking session. Update actual occupancy without counting the same allocation twice.	
Check Out a Private EV	Confirm the vehicle's departure, close the parking session, and release the occupied space. If charging is active, apply the agreed completion or interruption procedure before departure.	

IV. Operator Use Cases

Precondition: The operator is signed in

Fleet Monitoring and Redistribution Use Cases

Use case	Goal and expected outcome	Note
Monitor the Mobility Hub Network	View the network's vehicle distribution, parking utilization, charging availability, and operational problems. Identify Hubs approaching capacity limits or experiencing resource shortages.	
Inspect Vehicle Status	Inspect an individual vehicle's location, battery level, availability, rental state, and reported faults to support operational decisions.	
Inspect Charging-Point Status	Inspect a charging point's availability, operational condition, active session, and relevant scheduled usage.	
Coordinate Vehicle Redistribution	Select vehicles, source Hub, and destination Hub; check whether the transfer is feasible; and record a redistribution assignment. Creating the assignment does not mean that physical movement has already occurred.	Include: Validate Redistribution Feasibility - Check the selected vehicles and the destination's capacity before accepting the assignment.
Record Redistribution Completion	Confirm that assigned vehicles have reached their destination. Update their locations, assignment statuses, and source/destination occupancy consistently.	

Incident Handling Use Cases

Use case	Goal and expected outcome	Note
Handle an Operational Incident	General parent use case for investigating an incident, containing its effects, recording corrective action, and updating its resolution status.	
Handle a Vehicle Failure	Identify the affected vehicle, restrict unsafe or unavailable use, address affected reservations/rentals, and record recovery or repair status.	Specialization of above
Handle a Charging-Point Failure	Mark the point unavailable, identify affected charging sessions and allocations, initiate appropriate rescheduling, and record restoration.	Specialization of above

Smart Charging Use Cases

Use case	Goal and expected outcome	Note
View the Charging Queue and Schedule	Inspect pending requests, their priorities, allocated charging points, planned time intervals, and requests that cannot currently be scheduled.	
Generate a Charging Schedule	Ask the system to produce or refresh a schedule using battery levels, upcoming usage requirements, charging capacity, and resource availability. Return feasible allocations and identify unscheduled requests.	Include: Prioritize Charging Requests - Determine priorities using the agreed battery and upcoming-usage rules. Allocate Charging Slots - Assign feasible charging points and intervals without conflicting allocations.

Adjust a Charging Schedule	Modify permitted scheduling decisions, such as a charging interval or priority. Validate changes so that unavailable equipment and conflicting allocations are not accepted.	
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What-if Simulation Use Cases

Use case	Goal and expected outcome	Note
Configure a What-if Scenario	Define hypothetical conditions such as increased Metro arrivals, a full Hub, increased charging demand, charging-point failures, or excessive vehicle concentration. Specify the affected resources and simulation period.	
Run a What-if Simulation	Execute a configured scenario using a snapshot of the network state. Simulate resulting resource demand and state changes without modifying live operations.	Include: Obtain a Network Snapshot - Obtain the baseline state used for the simulation. Evaluate Scenario Impact - Calculate the scenario's effects on resources and service capacity. Generate Coordination Recommendations - Produce responses to the simulated conditions, including "no intervention required" where appropriate.
View Simulation Results	Inspect predicted effects on service capacity, parking utilization, vehicle availability, and charging congestion. Clearly distinguish simulated results from live status.	
Review Coordination	Inspect proposed responses,	

Recommendations	such as redistributing vehicles, adjusting charging schedules, or redirecting users to alternative Hubs, together with their expected effects.	
Compare What-if Scenarios	Compare multiple simulated scenarios or compare a scenario with a baseline using consistent evaluation measures.	

V. External Telemetry Provider Use Cases

Use case	Goal and expected outcome	Note
Update Vehicle Telemetry	Receive identified and timestamped vehicle data, such as battery level, location/Hub, and operational state; validate the update before accepting it.	
Update Parking Occupancy	Receive parking-space occupancy observations and reconcile actual occupancy with reservations and parking sessions.	
Update Charging-Point Status	Receive charging-point availability, fault, and session-status updates; update the operational state and identify affected allocations.	